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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: Data Science– (24AD3107)

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TITLE OF THE PROJECT	Smart Resume Screening and Job Recommendation System	
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Abstract:

The Smart Resume Screening and Job Recommendation System is an intelligent data science application developed to automate and improve the process of resume evaluation and job matching. Traditional recruitment processes require recruiters to manually review a large number of resumes, which can be time-consuming, inconsistent, and difficult to scale. This project addresses these challenges by applying Natural Language Processing (NLP) and recommendation techniques to analyze resumes and job descriptions automatically.

The system accepts a candidate's resume and a collection of job descriptions as input. It extracts relevant textual information such as technical skills, educational qualifications, work experience, projects, certifications, and job-related keywords. The extracted text is cleaned using NLP preprocessing techniques such as tokenization, stop-word removal, normalization, and stemming or lemmatization. The processed resume and job descriptions are then converted into numerical representations using TF-IDF (Term Frequency–Inverse Document Frequency).

To determine how closely a candidate matches a particular job, the system calculates Cosine Similarity between the resume and job-description vectors. Based on the calculated similarity scores, available jobs are ranked according to their relevance to the candidate's profile. In addition to job recommendation, the system performs skill-gap analysis by identifying required skills that are present in the job description but missing from the candidate's resume. This helps candidates understand which skills they should develop to improve their employment opportunities.

The system can also provide an interactive dashboard displaying match percentages, recommended jobs, matched skills, missing skills, and ranking results. By combining NLP, TF-IDF, similarity analysis, and recommendation techniques, the proposed system provides a practical solution for both job seekers and recruiters. It reduces manual screening effort, improves the efficiency of candidate-job matching, and enables more personalized and data-driven career recommendations.